Cisco PE Router QoS / Congestion Event Troubleshooting and Resolution SOP

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_3186

Description

This Standard Operating Procedure outlines the step-by-step process for identifying and resolving Critical QoS / Congestion Event failures on a Cisco PE router, specifically Node ID: HSR-PE-07 located at HSR Layout Access. The failure domain is characterized by Microbursts, interface tail drops, and QoS queue starvation, triggered by Alarm Code: ALARM_3186 with a Severity of Critical. The underlying protocol and hardware mechanics that contribute to this failure domain are rooted in the interaction between the Cisco PE router's queuing mechanisms, such as Weighted Random Early Detection (WRED), and the network's bandwidth allocation and policing policies. The router's ability to prioritize and manage traffic is further complicated by the presence of multiple service providers and network segments, which can lead to congestive microbursts if not properly configured.

Pre-requisite Checks

- Verify that remote hands are dispatched to the HSR Layout Access site to ensure physical access to the Cisco PE router.

- Check the status of redundant links to ensure that the primary link is operational and not contributing to the congestion event.

- Review the network topology and configuration to identify any potential bottlenecks or misconfigured QoS policies.

Potential Root Causes

Cause	Probability
Misconfigured WRED thresholds	High
Inadequate policing policies	Medium
Congestion on upstream network segments	Low

Diagnostic Commands & Expected Outputs

```
Command: show policy-map interface GigabitEthernet0/1, Expected Output: Input: packets 12345 bytes 67890 Output: packets 12345 bytes 67890, Description: This command will display the policy map configuration and statistics for the GigabitEthernet0/1 interface, which can help identify any misconfigured or inadequate policing policies.

Command: show queueing interface GigabitEthernet0/1, Expected Output: Interface GigabitEthernet0/1: Queueing strategy: Priority Queueing Maximum size: 1000 Current size: 500, Description: This command will
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display the queuing strategy, maximum size, and current size of the GigabitEthernet0/1 interface, which can help identify any issues with the router's queuing mechanisms.
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Resolution Steps

Step 1: Step: Configure WRED thresholds, Command: service-policy input policer-qos, Description: This command will apply the policer-qos policy map to the input interface, which will configure the WRED thresholds and help mitigate congestion.

Step 2: Step: Adjust policing policies, Command: policy-map policer-qos class class-default police 1000000 2000000 exceed-action policed-dscp-transmit, Description: This command will adjust the policing policies to ensure that traffic is not exceeding the configured rate, which can help prevent congestion.

Post-Check Verification

```
Command: show policy-map interface GigabitEthernet0/1, Expected Output: Input: packets 12345 bytes 67890 Output: packets 12345 bytes 67890, Description: This command will verify that the policy map configuration and statistics for the GigabitEthernet0/1 interface are as expected.

Command: show queueing interface GigabitEthernet0/1, Expected Output: Interface GigabitEthernet0/1: Queueing strategy: Priority Queueing Maximum size: 1000 Current size: 500, Description: This command will verify that the queuing strategy, maximum size, and current size of the GigabitEthernet0/1 interface are as expected.
```

Roll-back Procedures

- Step: Remove policer-qos policy map, Command: no service-policy input policer-qos, Description: This command will remove the policer-qos policy map from the input interface, which will restore the original configuration.

- Step: Restore original WRED thresholds, Command: service-policy input original-policer-qos, Description: This command will restore the original WRED thresholds and policing policies, which will ensure that the network returns to its previous state.